

1. (5.20)

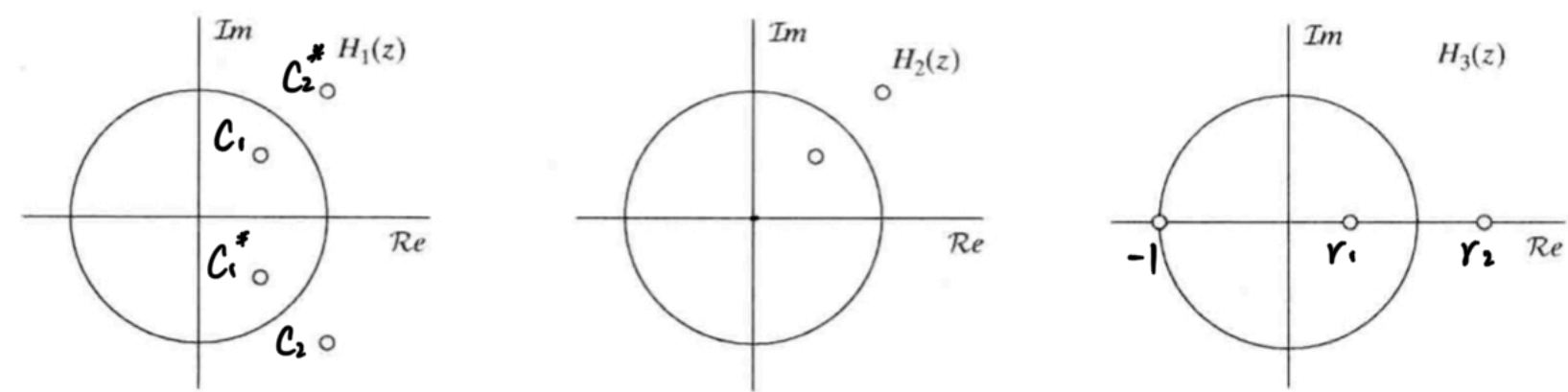


图 P5.20

$$\begin{aligned} (a) \quad H_1(z) &= (1 - c_1 z^{-1})(1 - c_2 z^{-1})(1 - c_1^* z^{-1})(1 - c_2^* z^{-1}) \quad [\text{能组成实系数线性差分方程}] \\ &= [1 - (c_1 + c_2)z^{-1} + c_1 c_2 z^{-2}][1 - (c_1 + c_2)^* z^{-1} + c_1^* c_2^* z^{-2}] \\ &= [z + c_1 c_2 z^{-1} - (c_1 + c_2)][z + c_1^* c_2^* z^{-1} - (c_1^* + c_2^*)] z^{-2} \end{aligned}$$

当 $c_1 = c_2^{-1}$ 时,

$$H_1(e^{j\omega}) = [2\cos\omega - (c_1 + c_2)][2\cos\omega - (c_1 + c_2)^*] e^{-j2\omega}$$

$$\angle H_1(e^{j\omega}) = -2\omega \Rightarrow H_1(z) \text{ 可能是广义线性系统}$$

(b) $H_2(z)$ 不存在共轭零极点 $\Rightarrow$ 不能组成实系数线性差分方程

(c) $H_3(z) = (1 - z^{-1})(1 - r_1 z^{-1})(1 - r_2 z^{-1}) \Rightarrow$ [能组成实系数线性差分方程]

$$= (1 - z^{-1})[1 - (r_1 + r_2)z^{-1} + r_1 r_2 z^{-2}]$$

$$= (z^{\frac{1}{2}} - z^{-\frac{1}{2}})[z + r_1 r_2 z^{-1} - (r_1 + r_2)] z^{-\frac{3}{2}}$$

$$\text{当 } r_1 = r_2^{-1} \text{ 时, } H_3(e^{j\omega}) = 2\sin(\frac{\omega}{2})[2\cos\omega - (r_1 + r_2)] e^{j(-\frac{3}{2}\omega + \frac{\pi}{2})}$$

$$\angle H_3(e^{j\omega}) = -\frac{3}{2}\omega + \frac{\pi}{2} \Rightarrow H_3(z) \text{ 是广义线性方程}$$

$$2. (5.33) \quad (a) \quad H(z) = \frac{(1 - 2z^{-1})(1 - 0.75z^{-1})}{z^{-1}(1 - 0.5z^{-1})} = -2 \cdot (1 - 0.75z^{-1}) \cdot \frac{z^{-1} - 0.5}{z^{-1}(1 - 0.5z^{-1})}$$

$$H_{\min 1}(z) = -2 \cdot (1 - 0.75z^{-1}), \quad H_{\text{ap}}(z) = \frac{z^{-1} - 0.5}{z^{-1}(1 - 0.5z^{-1})}$$

缩放尺度因子 -2, $H_{\min 1}(z)$ 和 $H_{\text{ap}}(z)$ 是唯一解

$$\text{因为 } H(z) = \frac{\prod_{|c_k| < 1} (1 - c_k z^{-1}) \prod_{|c_k| > 1} (1 - c_k^* z^{-1})}{\prod_{|d_k| < 1} (1 - d_k z^{-1}) \prod_{|d_k| > 1} (1 - d_k^* z^{-1})} = \underbrace{\frac{\prod_{|c_k| < 1} (1 - c_k z^{-1}) \prod_{|c_k| > 1} [1 - (c_k^*)^* z^{-1}]}{\prod_{|d_k| < 1} (1 - d_k z^{-1}) \prod_{|d_k| > 1} [1 - (d_k^*)^* z^{-1}]}}_{H_{\min 1}(z)} \cdot \underbrace{\prod_{|c_k| > 1} \frac{z^{-1} - c_k^{-1}}{1 - (c_k^*)^* z^{-1}} \cdot \prod_{|d_k| > 1} \frac{z^{-1} - d_k^*}{1 - d_k z^{-1}}}_{H_{\text{ap}}(z)} \cdot \underbrace{\prod_{|c_k| > 1} (-c_k) \cdot \prod_{|d_k| > 1} (-\frac{1}{d_k^*})}_{\text{Constant}}$$

所以分解唯一

$$(b) \quad H(z) = \frac{(1 - 2z^{-1})(1 - 0.75z^{-1})}{z^{-1}(1 - 0.5z^{-1})} = \frac{1 - 0.75z^{-1}}{(1 - 0.5z^{-1})^2} \cdot \frac{1 - 2z^{-1}}{z^{-1}} \cdot (1 - 0.5z^{-1}) = \frac{1 - 0.75z^{-1}}{(1 - 0.5z^{-1})^2} \cdot (z + z^{-1} - 2.5)$$

$$H_{\min 2}(z) = \frac{1 - 0.75z^{-1}}{(1 - 0.5z^{-1})^2}, \quad H_{\text{lp}}(z) = z + z^{-1} - 2.5 \quad [h[-1:1] = [1, -2.5, 1], \quad h[n] = 0 \text{ when } |n| > 1]$$

若算尺度因子, $H_{\min 2}(z) \rightarrow 2H_{\min 2}(z)$, $H_{\text{lp}}(z) \rightarrow \frac{1}{2}H_{\text{lp}}(z)$ 也是另一解

分解不唯一

$$3. (5.43) \quad (a) \quad H(z) = \frac{z^{-2}(1 - 2z^{-1})}{2(1 - \frac{1}{2}z^{-1})} = -\frac{z^{-3}(1 - \frac{1}{2}z^{-1})}{1 - \frac{1}{2}z^{-1}}$$

$$|H(e^{j\omega})| = |H(z)|_{z=e^{j\omega}} = 1 \Rightarrow H(z) \text{ 是全通系统}$$

$$(b) \quad H_{\min}(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \Rightarrow h_{\min}[n] = 2^{-n} u[n]$$

$$H_{\max}(z) = \frac{1 - 2z^{-1}}{2} \Rightarrow h_{\max}[n] = \frac{1}{2}(\delta[n] - 2\delta[n-1])$$

$$H_d(z) = z^{-2} \Rightarrow h_d[n] = \delta[n-2]$$

$$4. (5.52) \quad x[n] \rightarrow X(z)$$

$$\alpha^n x[n] \rightarrow X(\alpha^{-1}z)$$

$$\begin{aligned} X(\alpha^{-1}z) &= \frac{(1 - \frac{\alpha}{2}z^{-1})(1 - \frac{\alpha}{4}z^{-1})(1 - \frac{1}{6\alpha}z)}{1 - \frac{1}{6\alpha}z} \\ &= \frac{6}{5} \cdot \frac{(1 - \frac{\alpha}{2}z^{-1})(1 - \frac{\alpha}{4}z^{-1})(1 - 5\alpha z^{-1})}{1 - 6\alpha z^{-1}} \end{aligned}$$

要使 $X(\alpha^{-1}z)$ 是最小相位序列, 则要令

$$\max(\frac{|\alpha|}{2}, \frac{|\alpha|}{4}, 5|\alpha|, 6|\alpha|) < 1$$

$$\Rightarrow |\alpha| < \frac{1}{6}, \text{ 结合 } \alpha^n x[n] \in \mathbb{R}, \text{ 得}$$

即当 $-\frac{1}{6} < \alpha < \frac{1}{6}$ 时, $\alpha^n x[n]$ 满足题意

$$5. (5.61) \quad h_{\text{II}}[n] = h_{\text{II}}[M-n], \quad 0 \leq n \leq M, \quad M \text{ 是奇数}$$

$$H_{\text{II}}(e^{j\omega}) = A(e^{j\omega}) e^{j(-\frac{M}{2}\omega + \theta)}, \quad A(e^{j\omega}) \in \mathbb{R}$$

$$H(z) = H_{\text{II}}(1 - z^{-1}) \Rightarrow H(e^{j\omega}) = A(e^{j\omega}) e^{j(-\frac{M}{2}\omega + \theta)} \cdot (1 - e^{-j\omega})$$

$$= A(e^{j\omega}) \cdot 2\sin\frac{\omega}{2} e^{j(-\frac{M+1}{2}\omega + \theta + \frac{\pi}{2})} \Rightarrow \text{是广义线性相位系统}$$

$$H_{\text{II}}(z) \rightarrow h_{\text{II}}[n]$$

$$H_{\text{II}}(z)(1 - z^{-1}) \rightarrow h_{\text{II}}[n] - h_{\text{II}}[n-1] = h[n]$$

$$\text{当 } n < 0 \text{ 时, } h_{\text{II}}[n] = h_{\text{II}}[n-1] = 0 \Rightarrow h[n] = 0$$

$$\text{当 } n > M+1 \text{ 时, } h_{\text{II}}[n] = h_{\text{II}}[n-1] = 0 \Rightarrow h[n] = 0$$

$$\text{当 } 0 \leq n \leq M+1 \text{ 时, } h[M+1-n] = h[M+1-n] - h[M-n]$$

$$= h[n-1] - h[n] = -h[n] \text{ 是 III 类 FIR 广义线性相位系统。}$$